

DECODING TWITTER SENTIMENT FOR APPLE AND GOOGLE

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MORINGA SCHOOL

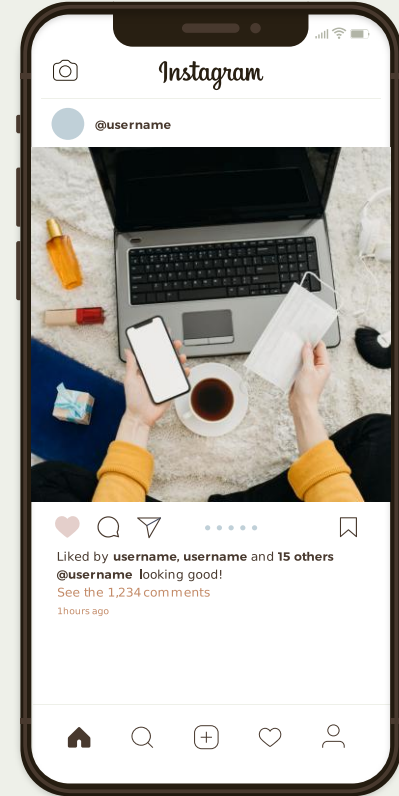




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OVERVIEW

- Organizations receive thousands of customer opinions daily, and manually analyzing the unstructured text is inefficient and unscalable.
- There is a need for an automated sentiment analysis model that accurately classifies tweets as positive, negative, or neutral to support data-driven decision-making and improve customer engagement strategies.
- To determine whether we can predict customer sentiment based on Tweets about sold products.

BUSINESS AND DATA UNDERSTANDING

- The dataset was obtained from data.world with 9,093 records and 3 features.
- Analysis was based on customer sentiments, i.e., positive, negative, or neutral, on Twitter about either Apple or Google.

CONT'

The processes undertaken on the data were:

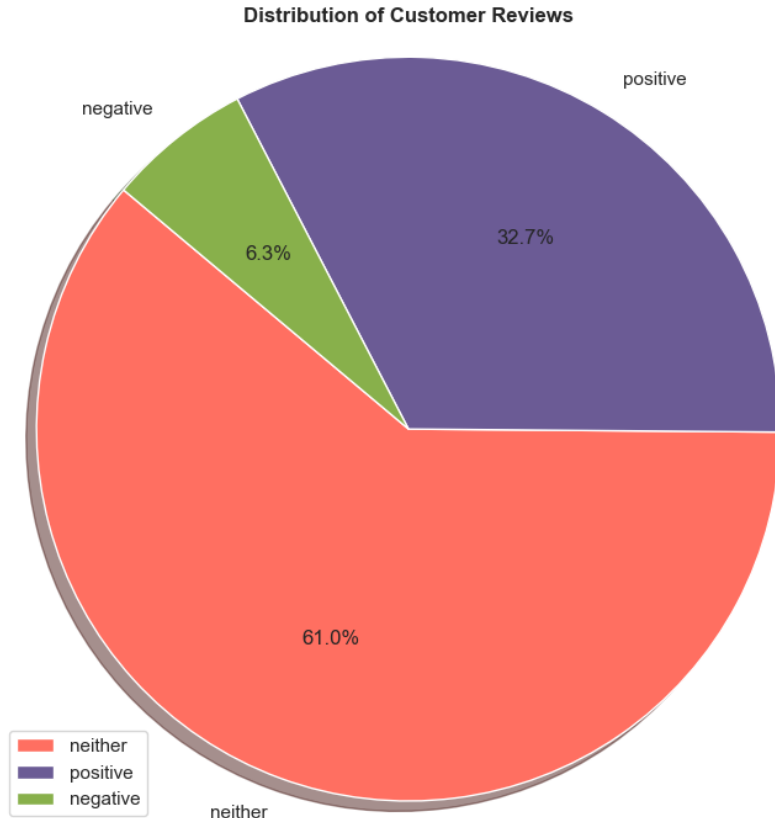
- Data cleaning(handling duplicates, missing values, and outliers)
- Removing stopwords, numbers, symbols, and punctuation.
- Tokenization of the words and Lemmatization.
- Splitting the data into train and test sets (75/25)
- Modelling.

Pipelines were used in the modeling to avoid data leakage.

STAKEHOLDERS

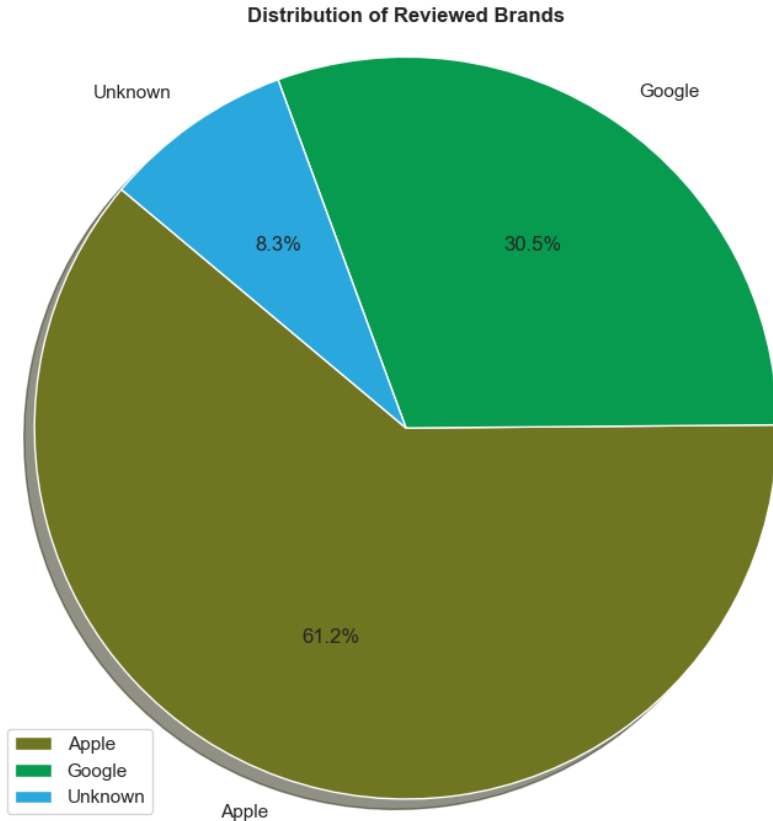
- Apple and Google Leadership and Executive.
- Product Development and Management Teams.
- Marketing and PR Departments.
- Customer Engagement and Support Teams.
- Social Media and Communications Managers.
- Event Organizers.

VISUALIZATIONS



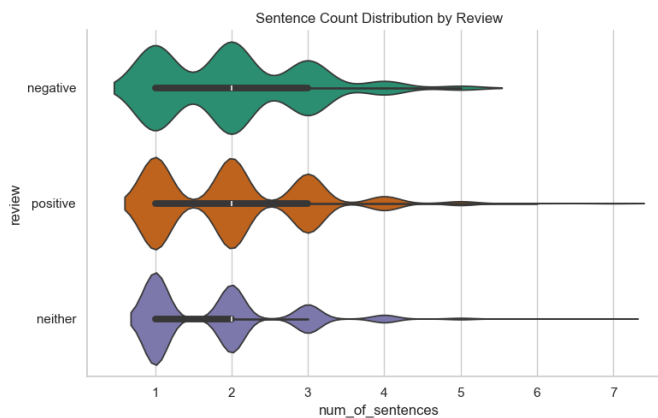
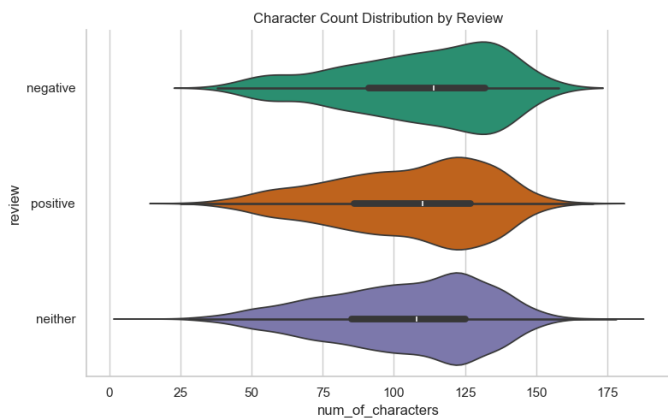
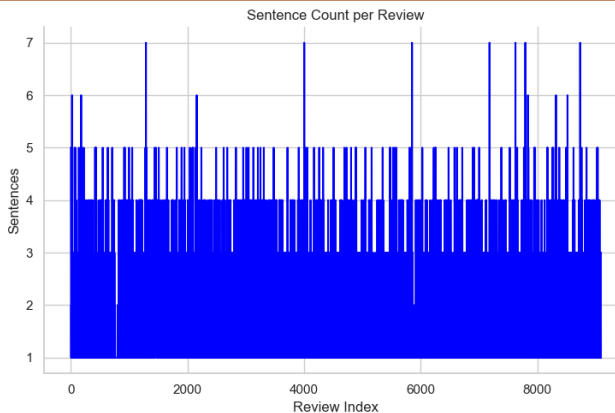
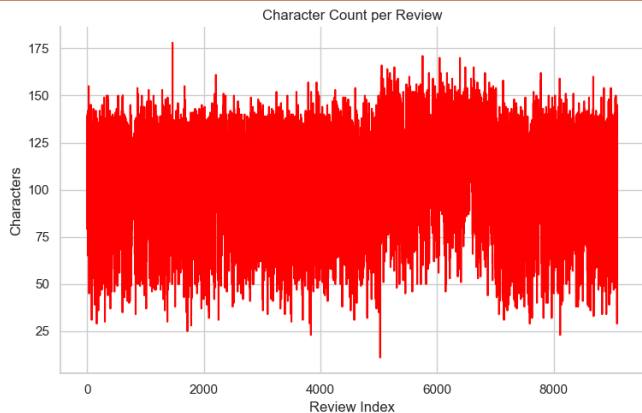
- The review distribution suggests that customer sentiment is largely neutral, with positive feedback significantly outweighing negative feedback.
- This could indicate that while many customers have moderate or mixed experiences, the overall perception is more favorable than unfavorable.

VISUALIZATIONS



- The reviewed brands are strongly skewed toward Apple, with Google being the second most frequently reviewed brand.
- Very few reviews could not be associated with a known brand.
- This suggests that the dataset is primarily focused on Apple and Google products or services.

VISUALIZATIONS



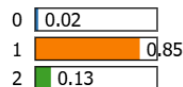
- Sentence count distributions overlap substantially across all sentiment categories.
- Therefore, sentence count is unlikely to be a highly discriminative feature for sentiment classification.

MODELING

- Eight models were developed: Random Forest as the baseline, a tuned Random Forest, Support Vector Machine(SVM), tuned SVM, MultinomialNB, a tuned MultinomialNB, K-nearest Neighbour, and a Keras-Sequential model.
- Across the eight models evaluated, accuracy ranged from 42% to 68%, with a clear split between the weak baseline and the rest of the tuned/alternative models.

EVALUATION

Prediction probabilities



NOT 0

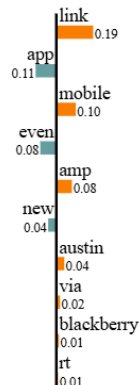
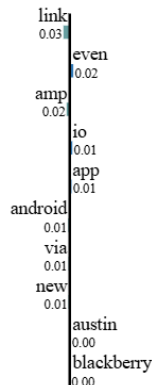
0

NOT 1

1

NOT 2

2



Text with highlighted words

rt link new mobile app store circular via android blackberry io even austin amp sxsw link

[('link', 0.19141271308345478), ('app', -0.1078263407067674), ('mobile', 0.10122353028034779), ('even', -0.08373619876115083), ('amp', 0.0808997130423862), ('new', -0.04244580680631921), ('austin', 0.04244580680631921), ('blackberry', 0.012345678901234567), ('via', 0.012345678901234567), ('rt', 0.012345678901234567)]

The lime plot above shows the model predicts Class 1 (85%), Class 2 (13%), and Class 0 (2%). Since the probabilities are varying (2% vs. 85% vs. 13%), the model's confidence in this prediction is relatively high.

INSIGHT AND CONCLUSION

- The developed sentiment classification model demonstrated the ability to automatically identify customer opinions expressed in tweets, providing a scalable alternative to manual sentiment assessment.
- The findings indicate that social media data can serve as a valuable source of information for understanding customer perceptions, monitoring brand reputation, and identifying emerging trends and concerns.

RECOMMENDATIONS

- Additional data from multiple social media platforms, such as Facebook, Reddit, and Instagram, could be incorporated to provide a more comprehensive view of customer sentiment.
- Aspect-based sentiment analysis could be performed to identify sentiments toward specific product features such as battery life, user interface, performance, pricing, and customer support.
- Businesses should integrate sentiment analysis dashboards into their decision-making processes to support marketing strategies, product improvements, and customer engagement initiatives.

THANK YOU

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